

Amir Reza Khatibi

Data & AI Engineer | Agentic AI, Machine Learning & Data Engineering

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Profile

Data & AI Engineer with hands-on experience building data pipelines, cloud platforms, and LLM-based AI systems. Skilled in **Python, SQL, LangGraph, FastAPI, dbt, Snowflake and AWS**, with experience in agentic workflows, ETL/ELT and data quality. Background in **Machine Learning and Deep Learning** with PyTorch, Transformers, transfer learning and fine-tuning, with a strong interest in reliable systems at the intersection of **Agentic AI, Machine Learning and Data Engineering**.

Experience & Teaching

Data Engineer Intern

05/2026 – Present

Target Reply

Turin, Italy

- Developed reusable data transformation models with **dbt** on **Snowflake** across multiple functional domains.
- Translated source mappings and project requirements into documented transformation logic and data entities.
- Used **Amazon S3** and **AWS Glue** in cloud-based data storage and processing workflows.

Teaching Assistant – Algorithms and Data Structures

08/2024 – 01/2025

Politecnico di Torino

Turin, Italy

- Supported over **200 students** with algorithms, data structures, complexity analysis, dynamic programming, graph traversal and debugging.

Technical Skills

AI & Agentic Systems	LangGraph, LangChain, LLM APIs, Tool Calling, Structured Outputs, Pydantic, LLM Guardrails, Context engineering, ReAct agents, Multi-agentic architectures
Machine Learning	PyTorch, Scikit-learn, Transformers, NLP, Computer Vision, RL, statistics
Data Engineering	SQL, Snowflake, dbt, PostgreSQL, PySpark, ETL/ELT, Data Modeling, Data Quality
Backend & Cloud	Python, FastAPI, REST APIs, Docker, AWS (EC2, S3, Glue), Railway
Engineering Tools	Git, Linux, GitHub Actions, pytest, Apache Airflow, Control-M

Research & Technical Projects

Autonomous Data Engineer Agent

github.com/Toukennn/Autonomous-data-engineer-agent

Live Demo – Railway

- Developed and deployed an **Agentic AI** platform with **LangGraph, LangChain and FastAPI**, using specialized agents for ETL workflows and SQL analytics over PostgreSQL and dbt.
- Designed deterministic guardrails for LLM-generated actions using **Pydantic**, AST validation with **SQLGlot**, relation allowlists and read-only database access.
- Implemented incremental API ingestion, schema evolution, data quality and lineage in a Medallion architecture, with **Docker**, automated tests, **GitHub Actions CI/CD** and cloud deployment on Railway.

EoMT Cross-Domain Semantic Segmentation and Anomaly Detection

github.com/Toukennn/eomt-cross-domain-segmentation

- Developed a **PyTorch** workflow based on the **Encoder-only Mask Transformer (EoMT)** for COCO–Cityscapes transfer learning, semantic segmentation and anomaly detection.
- Compared selective layer unfreezing with **LoRA-style parameter-efficient adaptation**; the best controlled fine-tuning run reached **75.3% validation mIoU**.

Weather ETL Pipeline

github.com/Toukennn/weather_ETL_pipeline

- Developed and deployed a **Python ETL pipeline** for REST API ingestion, transformation and storage on **AWS S3**, running on **AWS EC2**.
- Orchestrated scheduling, dependencies and retries with **Apache Airflow**, using Docker containerization and secure credential management through Airflow Variables.

Education

MSc in Data Science and Engineering

10/2025 – 10/2027

Politecnico di Torino

Turin, Italy

BSc in Computer Engineering

10/2022 – 10/2025

Politecnico di Torino

Turin, Italy

- Final grade: **110/110**; selected for the **INTRAPRENDENTI Project Honors School** and member of **IEEE-Eta Kappa Nu** (Honor Society).

Languages & Interests

Languages: Italian (B2), English (C1), Persian (C2 – Native), Turkish (A2)

Interests: Competitive chess, running, gym, learning new languages, reading